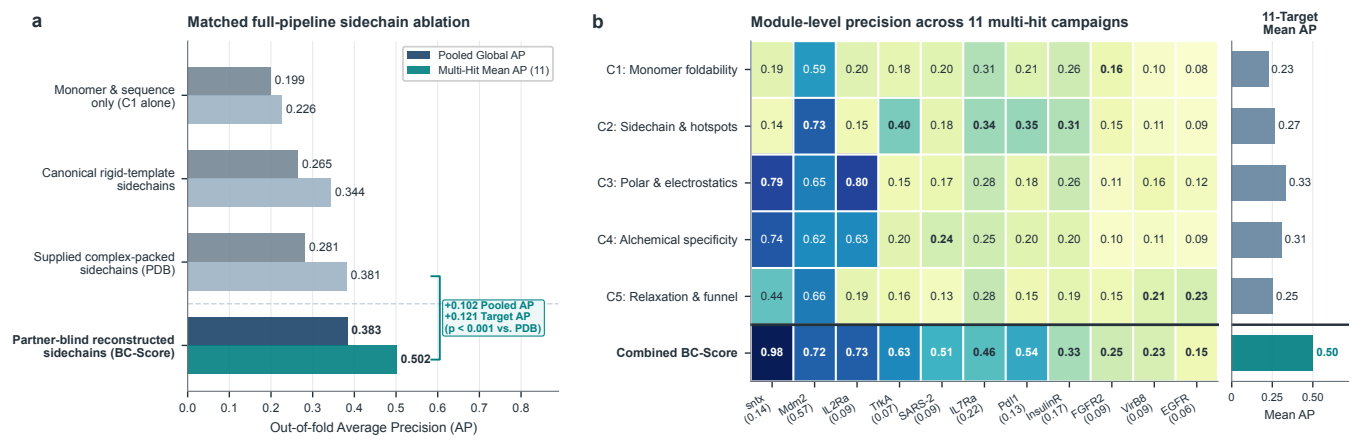

a
Matched full-pipeline sidechain ablation
Pooled Global AP
Multi-Hit Mean AP (11)
Monomer & sequence only (C1 alone)
0.199
0.226
Canonical rigid-template sidechains
0.265
0.344
Supplied complex-packed sidechains (PDB)
0.281
0.381
Partner-blind reconstructed sidechains (BC-Score)
0.383
0.502
+0.102 Pooled AP
+0.121 Target AP
(p < 0.001 vs. PDB)
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8
Out-of-fold Average Precision (AP)
b
Module-level precision across 11 multi-hit campaigns
11-Target Mean AP
C1: Monomer foldability 0.19 0.59 0.20 0.18 0.20 0.31 0.21 0.26 0.16 0.10 0.08
C2: Sidechain & hotspots 0.14 0.73 0.15 0.40 0.18 0.34 0.35 0.31 0.15 0.11 0.09
C3: Polar & electrostatics 0.79 0.65 0.80 0.15 0.17 0.28 0.18 0.26 0.11 0.16 0.12
C4: Alchemical specificity 0.74 0.62 0.63 0.20 0.24 0.25 0.20 0.20 0.10 0.11 0.09
C5: Relaxation & funnel 0.44 0.66 0.19 0.16 0.13 0.28 0.15 0.19 0.15 0.21 0.23
Combined BC-Score 0.98 0.72 0.73 0.63 0.51 0.46 0.54 0.33 0.25 0.23 0.15
0.23
0.27
0.33
0.31
0.25
0.50
srtx (0.14)
Mdm2 (0.57)
IL2Rα (0.09)
TrkA (0.07)
SARS-2 (0.09)
IL7Rα (0.22)
Pd-l1 (0.13)
InsulinR (0.17)
FGFR2 (0.09)
VirB8 (0.09)
EGFR (0.06)
0 0.25 0.50
Mean AP